World Development 109 (2018) 429–439

Contents lists available at ScienceDirect

World Development

journal homepage: www.elsevier.com/locate/worlddev

Sectoral Productivity Growth and Poverty Reduction: National and Global Impacts

Maros Ivanic [a], Will Martin [b]

[a] Independent Economist
[b] International Food Policy Research Institute, Washington, DC, USA

ARTICLE INFO

Article history:
Available online 18 August 2017

Key words:
total factor productivity
poverty
decomposition
sectoral growth
transformation

SUMMARY

This paper examines the implications of productivity improvements in agriculture, industry, and services for global poverty. We find that, in poor countries, increases in agricultural productivity generally have a larger poverty-reduction effect than increases in industry or services. This differential declines as average incomes rise, partly because agriculture becomes smaller as a share of the economy, and partly because agricultural productivity growth becomes less effective in reducing poverty. The source of the poverty-reduction benefits from agricultural productivity growth changes as innovations are more widely adopted—moving from increases in producer returns to reductions in consumer prices.
© 2017 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

A large body of research has demonstrated that the impacts of economic growth on poverty reduction depend heavily upon the sector in which the growth occurs. In particular, many studies (such as Ravallion & Datt, 1996) have found growth in rural areas to be associated with much more rapid reduction in poverty than economic growth in urban areas. This seems plausible given the generally much higher poverty rates in rural than in urban areas. However, if it is to be used as a basis for policy, it is important to understand the channels of effect so that policies can most effectively be targeted to promoting growth and poverty reduction.

Loayza and Raddatz (2010) make an important contribution by showing that the impact on poverty depends importantly upon the labor intensity of different sectors (e.g., agriculture, construction and services). In addition to this supply-side linkage, other studies have noted that widespread agricultural productivity growth may have important impacts by reducing the cost of food to poor consumers in closed economies or for the world as a whole (eg Dercon, 2009). On the other hand, Christiaensen, Demery, and Kuhl (2011) and Himanshu, Murgai, and Stern (2013) have concluded that the role of the non-agricultural sector in poverty reduction has increased, raising important questions about whether agriculture retains its traditionally central importance for poverty reduction.

Much of the available analysis of linkages between the sectoral composition of growth and poverty reduction outcomes has used econometric approaches. This has important advantages because it allows hypotheses to be tested against real-world data and confidence estimates to be assigned to parameter estimates. It also allows possible alternative channels of effect to be assessed and compared. Ravallion and Datt (1996), for example, found using econometric approaches that movement between sectors was less important for poverty reduction than growth rates within sectors.

Econometric approaches, however, face many challenges. The rate of output growth in each sector is clearly strongly endogenous, depending upon factors such as productivity growth rates, and yet it is common to use the output growth rate as an explanatory variable. While tests and potential solutions for endogeneity bias are widely used, the effectiveness of these approaches is uncertain given our inability to perform randomized control trials. It is also difficult to use econometric approaches to separate the direct impacts of productivity change through changes in producer incomes at a given commodity price, and those resulting from indirect impacts such as productivity-induced changes in wage rates and/or changes in commodity prices.

To complement the available econometric assessments and to allow assessments of global poverty impacts, we develop a framework that links a global Computable General Equilibrium (CGE) model with models for 315,000 households from 31 countries. This allows us to assess the poverty impacts of specific productivity shocks, and to take into account whether they apply just to one country or more broadly. This approach is based on rigorous use of economic theory and data on the structure of each economy and on patterns of international trade. For many of the questions that we consider—such as the impacts of changes in productivity—the key determinants of our results are actually the shares of income from different sources, and the shares of

http://dx.doi.org/10.1016/j.worlddev.2017.07.004
0305-750X/© 2017 The Authors. Published by Elsevier Ltd.
This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

expenditure on different goods, rather than elasticity parameters. The key parameters used—such as the elasticities of substitution between factor inputs, elasticities of consumer demand, and Armington elasticities of substitution between domestic and imported goods from different sources—are based on the econometric literature.

While one must always be aware of the potential limitations of using a simulation approach when the true structure generating the outcomes of interest is unknown, we are reassured by Kehoe's (2005) conclusion that simulation models are better suited to modeling productivity shocks than shocks such as trade reforms whose primary impact operates through price changes. Where results are available from both econometric studies and simulation models, and are broadly comparable, we can increase our confidence in both types of model. In this case, the ability of the simulation framework to compare different scenarios, such as technical change in a single, small economy facing fixed prices relative to the same change in a large set of countries, provides valuable additional insights. Where the results from simulation and econometric studies are not consistent, further analysis is needed to try to understand the differences.

Key questions addressed in this paper include: what is the relative effectiveness of productivity growth in agriculture, manufacturing and services in reducing poverty? In which countries do productivity increases have major impacts on global poverty? How are urban and rural poverty rates affected by productivity growth in different sectors? What are the sources of change in poverty rates? In the next section of this paper, we consider the specification of productivity changes, the impacts of productivity change on the prices of goods and factors, and the linkages between productivity growth and poverty. Then, in the third section, we consider the data and methodology used for the analysis. In the fourth section, we present and discuss key results. The final section presents conclusions.

2. Specifying productivity change

The first step in the causal chain from productivity to poverty is to identify the nature of the productivity growth under consideration. Uniform productivity growth[1] in all sectors is a special case and we consider scenarios in which productivity grows only in agriculture, in industry, or in services. Many different types of productivity growth might be considered, including productivity growth that augments different factors to different extents;[2] productivity growth that saves on intermediate inputs as well as on factors. Productivity growth may also be specific to particular geographical areas (this is frequently especially obvious in agriculture) or to particular types of firms. In this initial analysis, we focus on changes that augment all factors equally in production of a particular commodity. This lets us identify differences in poverty impacts that result from the two channels of effect identified in the literature—those resulting from differences in impacts on the earnings of the poor and on their cost of living.

Changes in productivity affect low-income households in three basic ways: (i) through changes in the productivity of the factors they employ in businesses, such as farms or service enterprises, that they operate; (ii) through changes in the prices of goods and services that they consume; and (iii) through changes in the factor returns (and particularly wages) that they receive from factors they sell outside their family-owned businesses.

Because of our concern about impacts on prices of goods and factors, we need a specification of productivity growth that takes into account all of the channels through which changes in productivity growth influence output. Obviously, improvements in productivity increase output directly by increasing the amount of output from any given level of inputs. However, another important channel of influence on output is through induced increases in profitability, for given input and output prices, that draw additional resources from competing activities. One way to capture both of these impacts of technological change is with the distinction between actual and effective units used in macroeconomics (Obstfeld & Rogoff, 1996, p431), in computable general equilibrium modeling (Dixon, 1982) and in the analysis of agricultural productivity growth (Martin & Alston, 1997). As we will see, this approach has the advantage of allowing a simple, graphical interpretation of the impacts of productivity change, as well as allowing a rigorous accounting for the full effects of productivity changes in technologies with multiple inputs and outputs.

With this approach, production can be represented using an unchanging profit function expressed in effective units of output and prices. From the point of view of the firm, quantity q^* of effective output now translates into a larger quantity, q, of output, where $q = q^*\tau$. The improvement in productivity also results in an increase in the effective price of output at any actual price, where the effective price is defined as $p^* = p\tau$. A quadratic profit function in effective prices provides a second-order approximation to any underlying technology and helps explain the impacts of this type of technical change:

$$\Pi = \alpha_0 + \alpha' p^* + \frac{1}{2} p^{*\prime} A p^* \tag{1}$$

where Π is the potential net return at current factor prices, p^* is as defined above and the α and A terms are coefficients.

Differentiating (1) with respect to the effective price yields a supply curve for the effective output of a particular sector.

$$q_i^* = \alpha_i + \sum_{ij} A_{ij} p_j^* \tag{2}$$

or

$$q_i = \tau_i \left(\alpha_i + \sum_{ij} A_{ij} p_j \tau_j \right) \tag{3}$$

Eqn. (3) can be depicted in actual price and quantity space as in Figure 1. As shown in this figure, productivity change has two effects on output at any given actual price. The first effect is an increase in output at any given input level associated with the τ_i term outside the parentheses in Eqn. (3). It increases output in the positive quadrant, and corresponds to the move from S_1 to S_2 in Figure 1. The second effect arises from the increase in profitability created by higher productivity, and is associated with the τ_i term within the parentheses on the right side of Eqn. (3). It changes the output (or input demand at points to the left of the vertical axis) at all prices above zero, and hence corresponds to the move from S_0 to S_1 in Figure 1. Note that this effect lowers the cutoff price at which positive quantities of output will be produced. As is clear from Eqn. (3), the move from S_1 to S_2 is a proportional change in output (from e to g in Figure 1) that is independent of the slope of the supply curve. By contrast, the increase in output associated with the rise in effective price (from f to e in Figure 1) depends upon the slope of the supply curve as well as the size of the technological change.

[1] For many purposes, it is useful to distinguish between improvements in technology and changes in the efficiency of firms relative to the technological frontier. We treat productivity growth from either source as equivalent in this initial analysis, although the differences between the two approaches could be explored in further analysis.

[2] The special case of labor-augmenting technical changes is widely used in macroeconomic models because it yields a balanced growth path. However, sectoral productivity growth may also augment land or capital, and growth paths may be far from balanced over substantial periods.

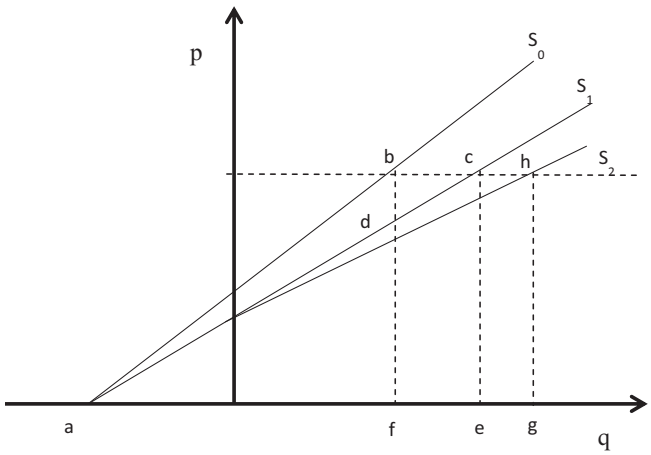

Figure 1. Impacts on output of an increase in productivity.

The first source of change in output—the increase in output at a given level of inputs—has a first-order impact on profits because it is "free." The second source of output increase comes about by attracting additional resources into the activity. Because the first-order impact of a productivity change tends to dominate the direct welfare impacts, it is frequently the only element considered in evaluations of the welfare impact of improvements in productivity (see, for example, Griliches, 1958). The second effect has only a second-order direct impact on profits at given prices, because of the cost of the additional inputs used to obtain this increase in output, but its effect on output, and hence on commodity prices, may be larger than the first-order impact. For our purposes, it is very important to distinguish between these two sources of increase in output following an increase in productivity. While, for some purposes, it may be reasonable to consider only the total effect on output, or only the direct impact on output, for welfare evaluation it is very important to distinguish the two sources of output change.

This exposition of technical change in terms of output bias is a very specific formulation, and productivity growth may take many forms, such as the changes biased toward saving particular inputs discussed in Bustos, Caprettini, and Ponticelli (2016). Our framework is general enough to allow all of these types of productivity growth to be analyzed if desired.

Where sectors are large, increases in productivity may change factor and commodity prices, which may have important impacts on the welfare of households and hence, potentially, on poverty. To capture these impacts in addition to the direct impacts considered above, we need to account for changes in the output prices facing poor and near-poor producers. In addition, we need to consider impacts on their costs of living, which are likely to be particularly important in the case of agriculture, given the large shares of expenditure devoted to food by the poorest consumers. Since many poor households obtain much of their income from sales of labor, we also need to account for impacts on wage rates. The procedures by which we accounted for these linkages are discussed in the next section.

3. Data and methodology

We use several key data sources for this work. First, we use the version 9 pre-release GTAP database (Aguiar, Narayanan, & McDougall, 2016) to support the GTAP 6.2 computable general equilibrium model. The database describes the economic relationships for 140 regions and 57 traded sectors of the world—including trade, production, and consumption for private households, governments and firms. In addition to the economic data, the database also provides a set of parameters describing trade, production, and consumption behavior.

Our second data source is household surveys from 31 countries, shown in Table 6. The resulting set of 315,000 households provides a fairly representative set of developing countries covering three quarters of the world's poor and all major pockets of poverty in the world. Most importantly it covers India, China, and most other large countries with large populations of poor people. All these surveys are from 2002 or later and include information not only on household consumption but also on agricultural production and income from sales of labor outside the household. As noted in Appendix Table 6, the vast majority of these surveys include information on households' business outputs and inputs in agriculture, industry and services. The surveys where some of this information is missing are relatively small in terms of population and their coverage of global poverty. This information allows us to understand and to model the important income sources and expenditure items for poor households, including their business income from non-agricultural sources.

The complexity of the task requires several distinct methodological steps in sequence. In the first step, we employ a global CGE model (GTAP) to identify the long-run implications of a set of stylized productivity shocks for national income, and for product and factor prices. In the second step, we apply the country-level productivity and price shocks to the 315,000 household-level models which we use to simulate the welfare, and hence the poverty, implications of the productivity shocks considered. Finally, we use the estimated sample properties of our country sample to estimate the poverty impacts at global and regional levels, and to place confidence bounds around these estimates.

4. The GTAP model

We use the GTAP model to estimate the price implications of the simulated productivity gains. Because this study is centered on very broad economic sectors and we have detailed household data on only a sub set of developing countries, we aggregate the 57 original sectors into three (agriculture, industry, and services) and the 140 GTAP regions into 40 (our 31 focus countries and other major economies such as the USA and the EU). Because productivity increases typically take some time, we implement a long-run closure which allows full mobility of capital and labor within each economy and partial mobility (with an elasticity of transformation equal to one) of land between agricultural activities while keeping other natural resources fixed.

Two types of labor are identified in each economy—skilled and unskilled—and these are assumed to be mobile between sectors but immobile internationally. We assume an unchanging level of unemployment in each region and allow wage rates to adjust in response to changes in supply and demand. In this mode, an increase in productivity in any one sector with output prices constant will likely draw resources away from other sectors because of the increase in profitability in the sector experiencing higher productivity. If output prices fall, consumers will benefit from lower living costs and other sectors will benefit from lower prices of goods used as intermediate inputs.

While, for most of our simulations, we use the standard GTAP model, we also consider a set of scenarios in which we simulate each country as a small, open economy. To do this, we fix the external prices facing the modeled country by allowing tariffs or export subsidies of those trading partners to adjust as needed to hold the domestic terms of trade constant. Because this breaks the requirement for global trade balance, it is necessary in this case to run the model separately for each simulated economy.

In each run of the (modified-where-necessary) GTAP model, we simulate improvements in total factor productivity for one of the sectors (agriculture and processed food, industry, services). The shock is either a 1% increase in productivity, which takes into account the declining importance of a one percent change in productivity in a sector that becomes smaller, or an increase equivalent to 1% of GDP—that is a shock that would increase value-added in the sector by 1% of initial GDP with input levels and prices unchanged. The second measure is actually more common in this literature, particularly in studies such as Ravallion and Datt (1996). It is complementary with the first in indicating the effectiveness of growth emanating from a particular sector in reducing poverty for a given increase in national output.

The GTAP model provides a set of changes of factor and goods prices along with estimates of effective producer output prices. Changes in effective prices for value added are converted into changes in effective output prices for the household models, which focus on output prices.

5. The household models

To understand the welfare and poverty measures reported in this study, it is helpful to use a dual specification for household behavior that includes: production responses within any unincorporated business activities; consumption behavior and factor supply by household members. This allows us to obtain a money measure of household welfare W at a given utility level, u, as:

$$\boldsymbol{W} = \pi(\boldsymbol{p}^*, \boldsymbol{w}, \tau) - \boldsymbol{e}(\boldsymbol{p}, \boldsymbol{w}, \boldsymbol{u}) = \boldsymbol{z}(\boldsymbol{p}, \boldsymbol{p}^*, \boldsymbol{w}, \tau, u). \tag{4}$$

where $\pi(\boldsymbol{p}^*, \boldsymbol{w}, \tau)$ is a profit function representing the profits generated by any unincorporated household enterprise, such as a farm firm, specified as a function of effective commodity prices, factor prices and technology; $e(\boldsymbol{p}, \mathbf{w}, u)$ is a full cost function of the type used by Deaton and Muellbauer (1981) for a household that consumes goods and supplies factors at a given vector of commodity prices, $\boldsymbol{p}$, factor prices, $\boldsymbol{w}$, and utility, u. Note that the actual prices of goods and factors are generally endogenous in the macro modeling stage but are always exogenous at the household level.

Eqn. (4) is general enough to encompass many different types of households. By duality, the vector of consumption for any household is given by $e_{\boldsymbol{p}}$. The vector of factors supplied by a household is similarly given by $-e_{\boldsymbol{w}}$. For specialized households, such as urban-wage-earning households, these two derivatives are sufficient to fully characterize household behavior. For a household with an unincorporated family business that uses some family labor and/or produces some portion of its consumption, net sales of commodities are given by $z_{\boldsymbol{p}}$ and net sales of factors are given by $z_{\boldsymbol{w}}$. Clearly, such households may be net buyers or net sellers of labor and commodities, with important implications for the welfare implications of price changes resulting from productivity changes. At the household level, we do not consider corporate enterprises, although they are included in the economy-wide analyses used to determine impacts on prices and factor returns.A second-order approximation of the welfare impact of changes in $\boldsymbol{p}$, w and τ may be compactly written for any type of household as:

$$\Delta W = [z_p z_w \pi_\tau] \begin{bmatrix} \Delta \boldsymbol{p} \\ \Delta w \\ \Delta \tau \end{bmatrix} + \frac{1}{2} [\Delta \boldsymbol{p} \quad \Delta w \quad \Delta \tau] \begin{bmatrix} z_{pp} & z_{pw} & \pi_{p\tau} \\ z_{wp} & z_{ww} & \pi_{w\tau} \\ \pi_{\tau p} & \pi_{\tau w} & \pi_{\tau\tau} \end{bmatrix} \begin{bmatrix} \Delta \boldsymbol{p} \\ \Delta w \\ \Delta \tau \end{bmatrix} \tag{5}$$

This quadratic form takes into account both the quadratic relationship between technological change variables and output levels in Eqn. (3) and the impacts of endogenous changes in quantities resulting from price changes and considered in Ivanic and Martin (2014). The first term in (5) includes the net sales of the household times the change in the price of the commodity, $z_p \Delta \boldsymbol{p}$, the measure of welfare change emphasized by Deaton (1989) for analysis of commodity price changes. It also takes into account the impact of changes in factor prices, and especially wage rates, times the net sales of the household outside its unincorporated enterprises such as farms, service or industrial enterprises, $z_w \Delta w$. Finally, it takes into account the direct impact of changes in technology on the profits generated by household firms given a change in technology, $\pi_\tau \Delta \tau$. As shown in Ivanic and Martin (2014), the second-order terms generalize these first-order impacts by adding induced changes in the output of household enterprises and changes in sales of labor when changes in technology and any resulting changes in prices are large.

Given our focus on productivity change, it is important to examine the direct impact of productivity change on the profits of family-owned firms in more detail than the other elements of this quadratic form, which are discussed in Ivanic and Martin (2014). The term for an individual good is readily seen to equal $\frac{d\pi}{dp^*} \frac{dp^*}{d\tau} = p.q^*$. The $\pi_{\tau\tau}$ term then equals $p.\frac{dq^*}{dp^*}.p$. Converting these into more familiar proportional-change form yields an intuitive expression for the second-order impact of a single technological change on household real income at constant nominal prices as:

$$\frac{\Delta \pi}{\pi} = \left[\frac{p_i^* q_i^*}{\pi}\right] . \frac{\Delta \tau_i}{\tau_i} + \frac{1}{2} \sum_j \left[\frac{p_j^* q_j^*}{\pi}\right] \eta_{ji} \left(\frac{\Delta \tau_i}{\tau_i}\right)^2 \tag{6}$$

where η_{ji} is the price elasticity of supply of good j with respect to the price of good i and all other terms are as previously defined. As shown in the discussion of Figure 1, this welfare effect consists of a first-order effect that depends only on the initial level of output, and a second-order impact that depends on the price responsiveness of output to price. As is evident from Eqn. (5), evaluation of Eqn. (5) requires estimates of the price responsiveness of household demands for consumption goods and of the output supply and input demand responses of household firms.

Once the estimated changes in real income for each household have been obtained, we count—by applying the available household weights—all households which move across the poverty line—defined at 1.25 USD/person/day or 2 USD/person/day—and calculate the corresponding change in the poverty headcount.

6. Representation of household demand responses

For consistency with the economy-wide model being used, we use the Constant Difference of Elasticities (CDE) specification to characterize consumer demand at the household level (Hertel, 1997). For consistency with the macro analysis, we use the estimated CDE substitution parameters from the GTAP model, together with each household's expenditure shares, to calculate its own- and cross-price elasticities of demand. Following Hanoch (1975, p. 409), we define a matrix of compensated elasticities ϵ for CDE preferences as:

$$\varepsilon_{ij} = \left(\alpha_i + \alpha_j - s^T \alpha - \frac{\delta_{ij} \alpha_i}{s_i}\right) s_j$$

where the $\alpha_i's$ are the CDE substitution parameters, s is a vector of consumption expenditure shares, and δ_{ij} is an indicator variable taking the value 1 when $i = j$ and 0 otherwise. The vector s for each household was obtained directly from the household survey data by calculating its consumption shares for each commodity. Because these expenditure shares vary by household, the matrix of elasticities is specific to each household. The parameter values, α, do not vary by household and were obtained from Reimer and Hertel (2004).

7. Representation of household firm input/output elasticities

Production of each good in the household firm model is represented by a two-level nested CES production function. In the bottom nest, the household firm combines its factors according to a Constant Elasticity of Substitution (CES) production function and then combines this value-added composite with material inputs using another CES production function to define total output. We make the default assumption that the CES elasticity between value added and intermediate inputs is zero, so that intermediate inputs are used in fixed proportion to output. Zero-profit conditions for each activity ensure that the price received for each output equals the cost of its production and define the supply price for each output. Household firms that produce multiple outputs are able to shift their factors between outputs with the same restrictions on factor mobility as are imposed on the national model. They are also able to transfer labor between household business activities and outside activities depending upon the returns available within the farm firm and outside.

For each household, we infer the input–output structure of any family firm using data from the relevant household survey and, where full details are not given in the survey, from the input–output structure for the relevant country or region. This allows us to create a set of input demand, zero profit and input use equations that we use to infer the set of long-run output supply and input demand elasticities needed to implement Eqn. (6). In the long-run closure used, all factors except land and natural resources are perfectly mobile between activities. Land and natural resources can be moved, but only when the return in use in the incoming sector rises relative to returns in its previous use, a situation which is represented very simply by allowing transformation of land between activities. Following the GTAP model an elasticity of transformation of one is applied. The household firm seeks to maximize its profits in producing its agricultural, industrial and services outputs. The elasticities of supply response take into account each household's pattern of production and consumption and its factor endowments. Because the initial input and output mix of each household is different, the elasticities of supply at the household level vary depending upon the range of products they produce. If, for instance, the share of rice in the output of a household is large, the household's elasticities of supply for rice will be relatively low. The resulting range of elasticities is given in Ivanic and Martin (2014).

8. Sample inference

The final step in our global estimation using poverty changes in our sample of 315,000 households to make assessments of global poverty impacts. This step includes applying sample weights to our national estimates to generate estimates of global poverty changes, and generating confidence bounds on these estimates. We extrapolate from sample-country poverty changes into estimated global poverty changes using the sample countries to represent other countries within the same World Bank region and income-group (low, lower middle and upper-middle income countries) to avoid sharp differences between sample and represented countries in structural characteristics such as labor shares, poverty rates and sectoral structure. Our sample covers most of the countries with substantial numbers of poor people, including the "pockets of poverty" among the lower middle income countries of South Asia and the upper middle-income countries of East Asia where our sample coverage is very high.

Since our sample covers 76% of the world's poor people, we were reasonably confident that changes in poverty based on this sample would provide reasonable approximations to changes in global poverty. To formalize this intuition, we used the weights for the countries included in our sample and for the 125 developing countries for which poverty rates were available to evaluate the standard errors associated with sample estimates. We then performed a Monte Carlo simulation of 10,000 runs with normally and independently distributed poverty deviations in the set of 125 developing countries, measured the differences between sample estimates and population outcomes, and calculated the variance of this difference. The ratio of this variance to the average weighted variance of the sample outcomes, was 2.6%, implying a standard error of sample estimates of 16% of the weighted standard deviation. The standard error for a shock, such as growth in any sectoral productivity, will depend on the standard deviation of responses to that shock.

9. Scenarios

We use a set of scenarios to assess the importance of improvements in total factor productivity in each sector for poverty reduction. In the first set of scenarios, we assume that each country whose productivity increases is a small open economy and hence productivity improvements translate only into changes in producers' effective prices and factor returns, and particularly wage rates for unskilled labor. In the second scenario, we model productivity change in each country using a standard GTAP model, which allows us to observe not only the implications of productivity shocks for wages and effective producer prices but also the implications for commodity prices. Finally, we simulate contemporaneous productivity gains, first in all developing countries, and then in all countries, which allow us to see how the world price implications of global productivity gains affect global poverty.

For all our scenarios, we consider both cases: (i) that directly increase the output of each sector by one percent at any given level of initial inputs, and (ii) that directly raise GDP by one percentage point. As we have seen, the actual change in output will be more than one percentage point at any given set of input and output prices because of the induced increase in resource use in the sector induced by higher productivity. The first experiment has an obvious interpretation in that technological improvements are typically assessed based on their impact on output at given inputs. The second experiment follows earlier studies such as Ravallion and Datt (1996). It is intended to provide an insight into the intensity of poverty reduction for productivity growth in a particular sector. If productivity rises in a sector that is more important as a source of income for the poor, or because declining prices following the productivity gain reduce living costs for the poor, then it may have a larger impact on poverty reduction than a productivity gain in a larger sector with weaker poverty-reduction linkages.

10. Results

The main results on the importance of productivity growth in different sectors for poverty reduction are presented in Table 1. In the first panel, Table 1(a), we show the change, in percentage points, in the poverty rate at $1.25 per day resulting from a one-percent increase in total factor productivity in agriculture, industry and services. In the second panel, Table 1(b), we raise productivity in each sector by one percentage point of GDP.

The first row of each set of results refers to the reduction in poverty when each country is considered as a small, open economy. The second row refers to the case where each country is considered individually, and affects the external prices that it faces—essentially, this is the apparent effectiveness of productivity growth to policy makers who are aware that productivity changes in their

Table 1
Global poverty impacts of total productivity shocks (at 1.25 USD/person/day), % points

Assumption	Agriculture	Industry	Services
(a) Productivity shock valued at one percent of GDP			
Small, open economies	−0.81	−1.35	0.02
Single country, GTAP	−0.87	−0.36	−0.26
Developing countries, GTAP	−0.89	−0.37	−0.27
Global productivity growth, GTAP	−0.88	−0.37	−0.30
(b) One-percent productivity shock			
Small open economies	−0.18	−0.22	−0.01
Single country, GTAP	−0.21	−0.08	−0.16
Developing countries, GTAP	−0.22	−0.07	−0.16
Global productivity growth, GTAP	−0.22	−0.06	−0.17

country affect the external prices that they face. The third row refers to the case where productivity change affects all developing countries—perhaps because it is an innovation developed and disseminated by an organization such as the Consultative Group on International Agricultural Research (CGIAR). The final row refers to a case where the innovation under consideration is adopted globally.

In both of these scenarios, it is clear that an increase in productivity has a bigger poverty-reducing impact if it arises in agriculture than in other sectors, with the exception of the first, open-economy scenario in which improvements in productivity in industry yield the largest poverty reduction. This result is due to the fact that several large countries in our sample, especially India and Pakistan, have larger increases in wage rates for unskilled workers when industrial productivity grows than when agricultural productivity grows[3]. Once output prices are allowed to change, however, these positive wage impacts diminish in importance as farm prices fall relative to prices of industrial products and poor consumers benefit.

The poverty impacts of improvements in agricultural productivity are surprisingly similar across the different scenarios, including the small, open economy case. In the small, open economy case, the productivity increase raises the incomes of farmers without reducing prices, and also raises factor prices, particularly the wage rate for the unskilled labor that is used intensively in developing country agriculture. Under the single-country scenario, the decline in output prices hurts producers and the factors used intensively in the now-more-productive sector, but it lowers the cost of living for consumers (including many small, poor farmers).

When we move to the impacts of a 1% increase in productivity in each sector shown in Table 1(b), agricultural productivity growth still makes a larger contribution to poverty reduction than either industry or services, but the margin is much smaller. Because of the generally larger size of the services sector and its extensive use as an intermediate input in the other sectors, its productivity growth has a larger impact on poverty than does industrial productivity growth.

11. What are the sources of changes in poverty rates?

Another key question is the source of the reductions in poverty. For this, we focus on the experiment in which productivity increases in each sector by enough to raise GDP by one percentage point. We perform a sequential decomposition of the poverty impacts where, in each step of the decomposition, we take into account only the incremental poverty impact of those price changes relative to the CPI that affect household welfare through changes in: (i) agricultural profits, (ii) other profits, (iii) the cost of food consumption, (iv) the cost of other consumption, and (v) wage earnings.

The results of the decomposition are presented in Table 2 for each of the four scenarios. For the small, open economy case with output prices constant, shown in Table 2(a), we find that most of the reduction in poverty from an increase in agricultural productivity arises from direct increases in agricultural profits; the corresponding wage implications are also favorable, albeit much smaller, and further contribute to reducing global poverty. For an increase in industrial productivity, most of the poverty reduction is transmitted through wage impacts with changes in profits being much less important. Because we assume that commodity prices remain constant in this small, open economy scenario, changes in productivity have no impact on the cost of living. Finally, a productivity increase in the service sector results in a small poverty increase with opposing impacts from rising service-sector profits in services firms owned by poor households and declining wages.

When we move, in Table 2(b), to the case where each country's productivity change has an impact on the prices it faces we notice that the reasons for reductions in poverty change. With rising agricultural productivity, the direct increases in profits are outweighed by the effects of lower agricultural output prices, an unsurprising result given inelastic consumer demands. Declining agricultural profits, in isolation, tend to increase poverty. On the other hand, the increase in relative output prices of other goods accompanying the reduction in food prices results in rising profits in other sectors and an associated reduction in poverty. While lower food prices hurt agricultural profits, they lower the cost of food, which contributes the most to the reduction in poverty observed in this scenario. Other prices increase relative to the CPI as wages rise, resulting in a small poverty increase due to more expensive non-food consumption.

When productivity rises in the industrial and service sectors, we observe a slightly different pattern of poverty reduction. Most importantly, the reductions in the cost of industrial and service inputs to agriculture relative to in agricultural prices result in noticeable poverty reductions through higher agricultural profits. The partiallyoffsetting reductions in output prices and the improvements in productivity result in slight reductions in poverty from changes in non-agricultural profits. While the rise in food prices relative to the CPI results in an increase in poverty, this is compensated for by a decline in the price of industrial output. In the case of services, which are less-consumed by the poor, the poverty reduction coming from lower service prices is too small to offset the poverty increase from higher food prices. However higher productivity in both industry and services increase wages, which contributes to poverty reduction.

As shown in Table 2(c), where all developing countries experience productivity improvements at the same time, the poverty implications always have the same signs as in the single-country experiments with endogenous prices, although the intensities are often different. With more countries improving their productivity, prices of the affected commodity fall more, which lowers consumer costs at the expense of producer profits. With higher agricultural productivity, each effect is amplified when all developing countries improve their productivity. In the case of industrial and service sector productivity, most of the poverty impacts are also amplified with the exception of "other producer profits" which decline, resulting in a smaller poverty reduction.

Where all countries experience productivity gains, as shown in Table 2(d), we observe similar intensification of the impacts on poverty. In the case of agricultural productivity, the biggest impact on poverty comes from declining consumer prices of

[3] This surprising result did not arise in earlier analysis based on the GTAP 8.2 database.

Table 2
Decomposition of poverty impacts of productivity shocks (at $1.25 USD/person/day), % points

Component	Agriculture	Industry	Services
(a) Small open economies			
Agricultural profits	−0.57	0.00	0.00
Other profits	0.01	−0.01	−0.11
Cost of food consumption	0.00	0.00	0.00
Cost of other consumption	0.00	0.00	0.00
Wage earnings	−0.25	−1.33	0.13
(b) Single Country, GTAP			
Agricultural profits	0.04	−0.11	−0.18
Other profits	−0.21	−0.05	−0.04
Cost of food consumption	−0.69	0.13	0.21
Cost of other consumption	0.28	−0.14	−0.04
Wage earnings	−0.29	−0.19	−0.21
(c) All developing countries			
Agricultural profits	0.15	−0.13	−0.19
Other profits	−0.29	−0.03	−0.04
Cost of food consumption	−0.78	0.17	0.23
Cost of other consumption	0.31	−0.18	−0.05
Wage earnings	−0.29	−0.19	−0.22
(d) All countries			
Agricultural profits	0.32	−0.19	−0.19
Other profits	−0.33	−0.02	−0.04
Cost of food consumption	−0.93	0.24	0.23
Cost of other consumption	0.37	−0.25	−0.08
Wage earnings	−0.29	−0.16	−0.23

food, which are particularly important for the poorest households. Increases in wage rates and, interestingly, increases in non-agricultural profits with declining costs of agricultural inputs are other important drivers of poverty reduction. For industrial and service sector productivity growth, falling costs of consumption goods are less important.

12. Does the sectoral intensity of poverty reduction vary with the poverty line?

Because Christiaensen *et al.* (2011) found differences between impacts on productivity at $1.25 per day and those at $2 per day at 2005 PPP, we repeated the analysis at the $2 per day poverty line. When we consider the poverty intensity measures for productivity shocks equal to one-percent of GDP, in Table 3, we find very similar results to those in the $1.25 per day case for each of the different assumptions considered in the model. In each case, we find that the agricultural sector continues to have much larger impacts on poverty reduction than either industry or services in all cases except the small, open economy case.

13. Relationship between sectoral productivity gains and country income

Another important question is the relationship between the poverty impact of technical change and the development level of the country. In countries with more important agricultural sectors, which are often poorer, we would expect improvements in agriculture to play a larger role in poverty reduction than similarlysized improvements in other sectors. We indeed see, in Figure 2 that poverty is reduced much more in the poorest countries by one-percent productivity increases[4] in agriculture than in industry or service sectors. We also observe that, with a few exceptions, the poverty impacts of improved productivity in services are larger than those in industry. As countries' per capita incomes rise, the importance of each type of productivity growth declines simply because the number of people near the international poverty line declines.

[4] Applied by individual countries in a general equilibrium setting.

Table 3
Global poverty impacts (at 2 USD/person/day) of productivity shock equal to 1% of GDP, % points

Assumption	Agriculture	Industry	Services
Small open economies	−1.00	−1.52	−0.04
Single country, GTAP	−0.82	−0.41	−0.31
Developing countries, GTAP	−0.85	−0.42	−0.32
Global simulation, GTAP	−0.90	−0.42	−0.33

The same holds true for the larger shocks equal to one percent of GDP where productivity improvements remain more important when applied to agriculture, as shown in Figure 3. As can be seen in Figure 3, the impact of the agricultural productivity increase remains greater than that for industry and services in almost all cases.

An improvement in productivity is generally expected to raise national income in the country that experiences it, except in cases such as those considered by Bhagwati (1956) where a country's market power in trade is not internalized or by Johnson (1967) where the productivity growth occurs in sectors that are strongly subsidized. However, it is not necessarily true that it will always be poverty reducing, even in cases where these conditions are not met. Consider, for example, the case of Ecuador where an improvement in productivity often results in increases in poverty, especially in the case of an improvement to agricultural productivity equal to one percent of the country's GDP, with poverty increasing by 2.07% points as a result (see Table 4). This paradoxical result is caused by the fact that higher productivity in agriculture raises wages relative to the CPI, reducing the non-agricultural profits of labor-intensive non-agricultural businesses run by poor people.

We find very similar results when we repeat the exercise with a global simulation with the same productivity shocks to all countries at the same time. As can be seen from Figures 4 and 5, the importance of productivity increases in agriculture for poverty reduction generally exceed those for industry and services. Again, however, poverty reductions decline with rising per-capita incomes, regardless of which sector experiences the improvement in productivity.

14. Which countries' productivities have the largest impacts on global poverty?

An important policy question for any estimate of global poverty impacts is which countries contribute the most to the overall poverty outcome. This is also important for the reliability of the estimates. If they depend almost entirely on the results for one or two countries, this is a particular concern because of potential problems with the design of the survey instrument, our interpretation of it, or many other potential sources of error. To understand the role of individual countries on the global results, we calculate the impacts of individual countries on the global estimate. We report the results for a wide range of scenarios in Table 7, showing the mean poverty impact, the standard error of that estimate, and the contribution to the total of the three most influential countries.

The general finding from this table is that the standard errors of the estimates are generally small to very small relative to the mean estimates. The three most influential countries in the results are usually China, India and one other country such as Bangladesh, Indonesia, Nigeria or Pakistan. Excluding China or India would generally change the result noticeably but not lead to a qualitative change in the results. If we focus, for instance, on the impacts of

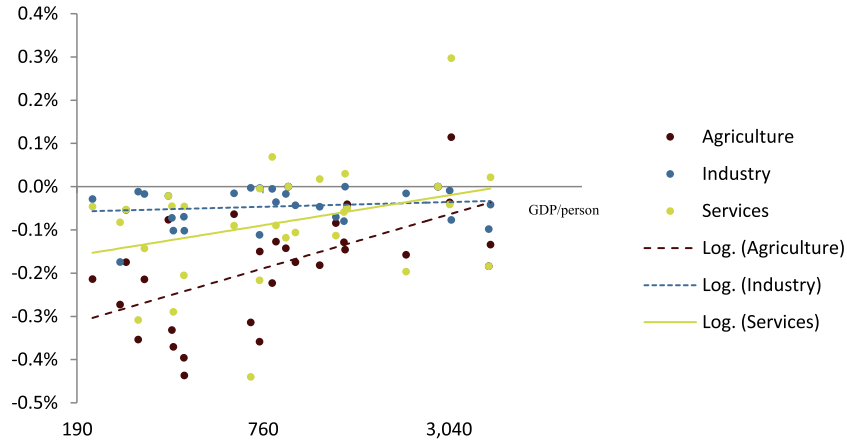

Figure 2. Relationship between per capita GDP and poverty change from a 1% productivity change (single country simulations).

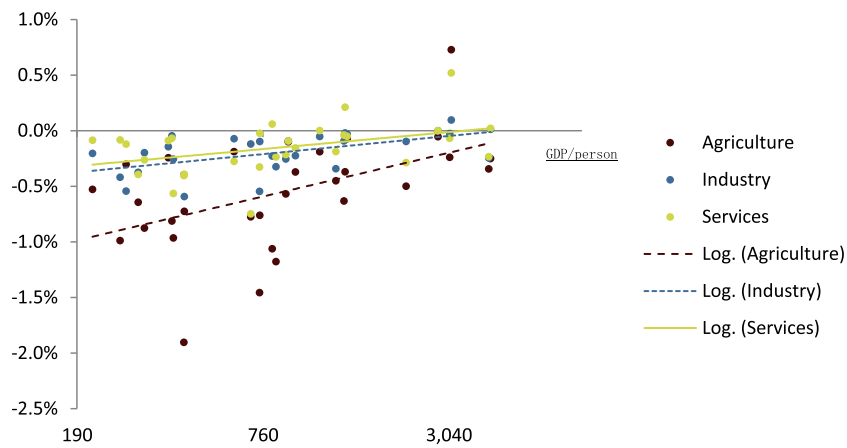

Figure 3. Relationship between per capita GDP and poverty change from a productivity increase equal to 1% of GDP (single country simulations).

Table 4
Poverty implications (at 1.25 USD/person/day) of productivity gains in different sectors applied by all countries, GE simulation, in percentage points

Country	GDP per capita	Agriculture (1 pct)	Industry (1 pct)	Services (1 pct)	Agriculture (1% of GDP)	Industry (1% of GDP)	Services (1% of GDP)
Malawi	213	−0.19	−0.04	−0.04	−0.40	−0.25	−0.09
Niger	262	−0.19	−0.25	−0.08	−0.50	−0.40	−0.23
Rwanda	274	−0.20	−0.08	−0.07	−0.30	−0.58	−0.17
Nepal	300	−0.39	0.02	−0.31	−0.60	−0.37	−0.38
Uganda	314	−0.15	−0.04	−0.13	−0.87	−0.32	−0.26
Tajikistan	375	−0.07	−0.02	−0.02	−0.13	−0.24	−0.09
Sierra Leone	385	−0.27	−0.01	−0.05	−1.07	0.09	0.00
Cambodia	389	−0.35	−0.13	−0.36	−0.72	−0.39	−0.65
Bangladesh	421	−0.51	0.02	−0.22	−2.61	−0.20	−0.43
Tanzania	422	−0.25	−0.25	−0.16	−0.34	−0.94	−0.44
Nigeria	612	−0.06	0.00	−0.09	0.00	−0.11	−0.27
Pakistan	693	−0.31	0.01	−0.46	−0.82	−0.06	−0.83
India	740	−0.39	−0.06	−0.22	−1.60	−0.50	−0.31
Zambia	741	−0.16	0.02	−0.02	−0.84	−0.05	−0.04
Mongolia	813	−0.22	−0.01	0.05	−1.12	−0.16	0.03
Yemen	837	−0.16	−0.03	−0.12	−1.19	−0.44	−0.24
Viet Nam	900	−0.09	−0.10	−0.14	−0.33	−0.23	−0.27
Moldova	917	0.00	0.00	−0.09	−0.02	−0.09	−0.09
Cote D'Ivoire	967	−0.05	−0.09	−0.08	0.67	−0.39	−0.21
Nicaragua	1,159	−0.04	0.00	0.04	0.13	0.18	0.09
China	1,307	−0.08	−0.10	−0.14	−0.29	−0.45	−0.29
Indonesia	1,389	−0.14	−0.03	−0.06	−0.91	−0.01	−0.03
Sri Lanka	1,400	−0.15	0.03	0.03	−0.20	0.15	0.17
Armenia	1,422	−0.04	−0.05	−0.05	−0.07	−0.03	−0.05
Guatemala	2,207	−0.11	−0.01	−0.18	−0.52	−0.04	−0.30
Albania	2,799	0.00	0.00	0.00	0.03	0.00	0.00
Peru	3,054	−0.05	−0.01	−0.04	−0.27	−0.01	−0.07
Ecuador	3,090	0.27	−0.05	0.30	2.07	0.27	0.51
Belize	4,086	−0.18	−0.18	−0.18	−0.26	−0.55	−0.23
Panama	4,140	−0.09	0.00	0.02	−0.37	−0.03	0.00

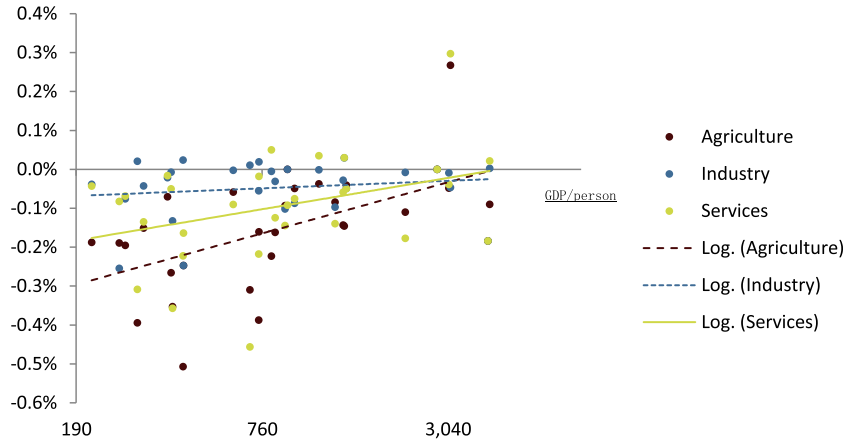

Figure 4. Relationship between per capita GDP and poverty change from a 1% productivity increase (global simulation).

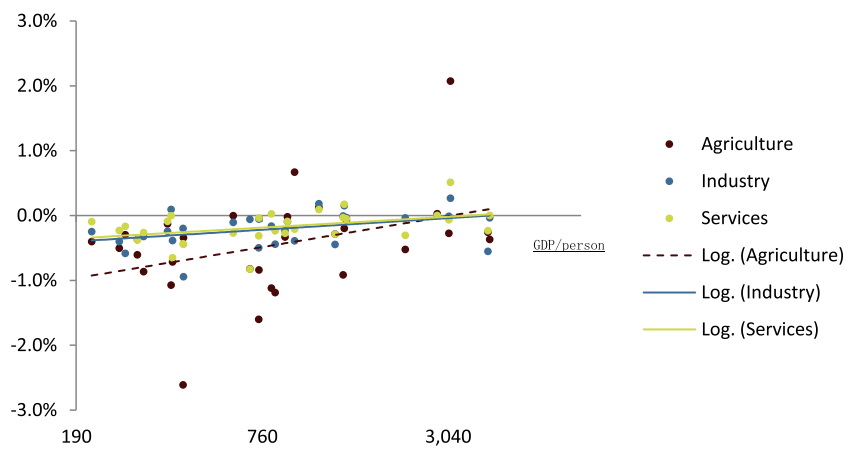

Figure 5. Relationship between per capita GDP and poverty change coming from productivity change equal to 1% of GDP (global simulation).

a 1% increase in output coming from agriculture, the overall impact is −0.22 with a standard error of 0.02. Exclusion of India would increase the estimate by 0.05% points, while the exclusion of both China and India would leave the result unaffected. We observe a similar pattern in other simulations which suggests that our global poverty estimates are not driven exclusively by a very small number of countries but reflect broadly representative effects within the developing world.

15. How are urban and rural poverty rates affected?

While the impacts of productivity increases on poverty do not appear to vary greatly depending whether the economy is open or closed (or whether the increase is global), the impacts on urban and rural households vary considerably. As is evident from Table 5 (a) the impact of higher agricultural productivity on rural poverty is highest in the small, open economy case where agricultural prices do not decline in response to the increase in productivity. This favorable impact declines modestly as we move to cases, such as the global TFP case, where agricultural output prices decline.

When we move to the urban poverty impacts in Table 5(b), we see that increases in productivity in the industry sector have the largest impact on poverty, because of the importance of this sector in the incomes of the urban poor. As we move to the scenarios where prices decline, agricultural productivity growth contributes the most to urban poverty reduction because of the importance of food prices for the real incomes of the poor.

Table 5
Global poverty impacts of total productivity shock equal to 1% of GDP (at 1.25 USD/person/day), % points

Assumption	Agriculture	Industry	Services
(a) Rural			
Small open economies	−1.04	−1.41	0.02
Single country, GTAP	−0.98	−0.42	−0.28
Developing countries, GTAP	−0.97	−0.44	−0.29
Global simulation, GTAP	−0.92	−0.45	−0.33
(b) Urban			
Small open economies	−0.29	−1.21	0.01
Single country, GTAP	−0.63	−0.23	−0.22
Developing countries, GTAP	−0.70	−0.21	−0.22
Global simulation, GTAP	−0.77	−0.19	−0.23

16. Conclusions

The key finding of this study work is that productivity gains in agriculture are generally, but not always, more effective in reducing global poverty than equivalent-sized productivity gains in industry or services. This means that increasing GDP by identical amounts through increasing productivity in different sectors

would lower poverty most if the productivity gain is in agriculture. Even when the size of the productivity gain in agriculture is not adjusted for the lower share of agriculture in the majority of developing countries, poverty reductions through improvements in agriculture are frequently on par with the gains from equally sized productivity gains in all other sectors. These results are consistent with the findings from the econometric studies by Loayza and Raddatz (2010) and Ravallion and Datt (1996).

We also show that the poverty reductions from agricultural productivity growth are typically not very responsive to the source of the welfare benefits of the productivity shock. Countries appear to benefit roughly equally in terms of poverty reduction no matter whether they implement productivity gains alone or as part of larger groups of countries implementing at the same time. However, even though the overall reductions in poverty are largely unaffected, there are substantial differences in the implications of agricultural productivity growth for urban and rural households—when the productivity gain is localized it reduces output prices less which benefits agricultural households; when the gain is more widespread, output prices decline more, which tends to benefit urban households more than farming households.

In this work, we believe we were able to estimate the global implications of productivity scenarios with reasonable confidence for two main reasons. First, we derived our results from a large and representative sample of developing countries which included key countries, such as China and India, with large numbers of poor people. Secondly, we find that the implications of productivity improvements for poverty are generally broadly similar across most developing countries. This means that exclusion or addition of even the largest developing countries from the simulated scenarios does not usually affect the sign of the global poverty change, although it does change the magnitude of the global poverty impact because of the size of the developing country in question.

Interestingly, we find that the impacts of productivity gains for poverty rates in particular countries are generally not very responsive to whether external prices change or how many developing countries take part in the productivity improvements—when only single countries improve their productivity, their poverty falls as a result of profit gains to producers; when more countries improve their productivity, poverty falls at about the same rate. This is particularly marked for agricultural productivity growth, where small, poor farmers are frequently net food buyers spending large shares of their income on food.

Appendix A

Table 6
Surveys used in this study

Country	Year	Survey name	Number of households	Agricultural profits	Agricultural inputs	Industry profits	Industry inputs	Services profits	Service inputs
Albania	2005	Living Standards Measurement Survey	3,664	Yes	Yes	Yes	Yes	Yes	Yes
Armenia	2004	Integrated Survey of Living Standards	6,815	Yes	Yes	Yes	Yes	Yes	Yes
Bangladesh	2005	Household Income-Expenditure Survey	10,080	Yes	Yes	Yes	Yes	Yes	Yes
Belize	2009	Household Income and Expenditure Survey	1,948	Yes	Yes	Yes	Yes	No	Yes
Cambodia	2003	Household Socio-economic Survey	14,984	Yes	Yes	No	Yes	Yes	Yes
China	2002	Chinese Household Income Project	5,783	Yes	Yes	Yes	Yes	Yes	Yes
Cote D'Ivoire	2002	Enquete Niveau de Vie des Menages	10,798	Yes	Yes	Yes	Yes	Yes	Yes
Ecuador	2006	Encuesta Condiciones de vida – Quinta Ronda	13,581	Yes	Yes	Yes	Yes	Yes	Yes
Guatemala	2006	Encuesta Nacional de Condiciones de Vida	13,686	Yes	Yes	Yes	Yes	Yes	Yes
India	2005	India Human Development Survey (IHDS)	41,554	Yes	Yes	Yes	Yes	Yes	Yes
Indonesia	2007	Indonesia Family Life Survey	12,999	Yes	Yes	Yes	Yes	Yes	Yes
Malawi	2004	Second Integrated Household Survey	11,280	Yes	Yes	Yes	Yes	Yes	Yes
Moldova	2009	Cercetarea Bugetelor de Familie	5,532	Yes	Yes	No	Yes	No	Yes
Mongolia	2002	Household Income and Expenditure Survey	3,308	Yes	Yes	Yes	Yes	Yes	Yes
Nepal	2002	Nepal Living Standards Survey II	5,071	Yes	Yes	Yes	Yes	Yes	Yes
Nicaragua	2005	Encuesta Nacional de Hogares sore Medicion de Nivel de Vida	6,619	Yes	Yes	Yes	Yes	Yes	Yes
Niger	2007	Enquete National sur Le Budget et la Consommation des Menages	4,000	Yes	Yes	Yes	Yes	Yes	Yes
Nigeria	2003	Nigeria Living Standards Survey	19,121	Yes	Yes	Yes	Yes	Yes	Yes
Pakistan	2005	Pakistan Social and Living Standards Measurement Survey	15,453	Yes	Yes	Yes	Yes	Yes	Yes
Panama	2003	Encuesta de Niveles de Vida	6,362	Yes	Yes	Yes	Yes	Yes	Yes
Peru	2007	Encuesta Nacional de Hogares	22,201	Yes	Yes	Yes	Yes	Yes	Yes
Rwanda	2005	Integrated Household Living Conditions Survey	6,900	Yes	Yes	Yes	Yes	Yes	Yes
Sierra Leone	2011	Sierra Leone Integrated Household Survey	6,737	Yes	Yes	Yes	Yes	No	No
Sri Lanka	2007	Household Income and Expenditure Survey	4,633	Yes	Yes	Yes	Yes	Yes	Yes
Tajikistan	2007	Living Standards Measurement Survey	4,644	Yes	Yes	No	Yes	Yes	Yes
Tanzania	2008	National Panel Survey	3,264	Yes	Yes	Yes	Yes	Yes	Yes
Timor-Leste	2007	Poverty Assessment Project	4,477	Yes	Yes	Yes	Yes	Yes	Yes
Uganda	2005	Socio-Economic Survey	7,425	Yes	Yes	Yes	Yes	Yes	Yes
Viet Nam	2010	Household Living Standard Survey	9,399	Yes	Yes	Yes	Yes	Yes	Yes
Yemen	2006	Household Budget Survey	13,136	Yes	Yes	Yes	Yes	Yes	Yes
Zambia	2010	Living Conditions Monitoring Survey	19,398	Yes	Yes	Yes	Yes	Yes	Yes

Table 7
Total average poverty implications (at 1.25 USD/person/day), in percentage points (standard errors in parentheses, top contributors to the global estimate shown)

Scenario	Agriculture	Industry	Services
1% (of GDP) shock, developing countries	Global change: −0.89 (0.10) Top contributors: India −0.21, China +0.17, Bangladesh −0.06	Global change: −0.37 (0.03) Top contributors: India −0.05, Indonesia +0.02, Pakistan +0.01	Global change: −0.27 (0.02) Top contributors: Pakistan −0.02, China +0.02, India −0.02
1% (of GDP) shock, global simulation	Global change: −0.88 (0.12) Top contributors: India −0.24, China +0.21, Bangladesh −0.07	Global change: −0.37 (0.03) Top contributors: India −0.04, China −0.03, Indonesia +0.02	Global change: −0.30 (0.03) Top contributors: Pakistan −0.02, Indonesia +0.02, Bangladesh −0.01
1% (of GDP) shock, single country simulation	Global change: −0.87 (0.08) Top contributors: India −0.19, China +0.15, Bangladesh −0.04	Global change: −0.36 (0.03) Top contributors: India −0.06, Indonesia +0.02, Nigeria +0.01	Global change: −0.26 (0.03) Top contributors: China +0.03, Pakistan −0.02, India −0.02
1% (of GDP) shock, single country simulation (SOE)	Global change: −0.81 (0.05) Top contributors: China −0.09, India +0.03, Indonesia +0.03	Global change: −1.35 (0.29) Top contributors: China +0.33, India −0.32, Pakistan −0.32	Global change: 0.02 (0.05) Top contributors: Pakistan +0.04, India −0.03, China +0.03
1 +% shock, developing countries	Global change: −0.22 (0.02) Top contributors: India −0.05, China +0.05, Bangladesh −0.01	Global change: −0.07 (0.01) Top contributors: China −0.01	Global change: −0.16 (0.02) Top contributors: India −0.02, China +0.02, Pakistan −0.01
1% shock, global simulation	Global change: −0.22 (0.02) Top contributors: India −0.05, China +0.05, Bangladesh −0.01	Global change: −0.06 (0.01) Top contributors: China −0.01	Global change: −0.17 (0.01) Top contributors: India −0.01, Pakistan −0.01, China +0.01
1% shock, single country simulation	Global change: −0.21 (0.02) Top contributors: India −0.05, China +0.05, Bangladesh −0.01	Global change: −0.08 (0.01) Top contributors: India −0.01	Global change: −0.16 (0.02) Top contributors: India −0.02, China +0.02, Pakistan −0.01
1% shock, single country simulation (SOE)	Global change: −0.18 (0.01) Top contributors: Indonesia +0.01, China +0.01	Global change: −0.22 (0.04) Top contributors: India −0.05, China +0.05, Pakistan −0.04	Global change: −0.01 (0.03) Top contributors: Pakistan +0.02, China +0.02, India −0.02

References

Aguiar, A., Narayanan, B., & McDougall, R. (2016). An overview of the GTAP 9 Data Base. *Journal of Global Economic Analysis, 1*(1), 181–208 https://jgea.org/resources/jgea/ojs/index.php/jgea/article/view/23.

Bhagwati, J. (1956). Immiserizing growth: A geometrical note. *Review of Economic Studies, 25*(3), 201–205.

Bustos, P., Caprettini, B., & Ponticelli, J. (2016). Agricultural productivity and structural transformation: Evidence from Brazil. *American Economic Review, 106*(6), 1320–1365.

Christiaensen, L., Demery, L., & Kuhl, J. (2011). The (evolving) role of agriculture in poverty reduction—An empirical perspective. *Journal of Development Economics, 96*, 239–254.

Deaton, A. (1989). Rice prices and income distribution in Thailand: A non-parametric analysis. *The Economic Journal, 99*(395), 1–37.

Deaton, A., & Muellbauer, J. (1981). Functional forms for labor supply and commodity demands with and without quantity restrictions. *Econometrica, 49*(6), 1521–1532.

Dercon, S. (2009). Rural poverty: Old challenges in new contexts. *World Bank Research Observer, 24*(1), 1–28.

Dixon, P. (1982). *ORANI, a Multisectoral Model of the Australian Economy, North-Holland, New York*.

Griliches, Z. (1958). Research costs and social returns: Hybrid corn and related innovations. *Journal of Political Economy, 66*(5), 419–431.

Hanoch, G. (1975). 'Production and demand models with direct or indirect implicit additivity. Econometrica, *43* (1), 395–419.

Hertel, T. (1997). *Global Trade Analysis*. Cambridge: Cambridge University Press.

Himanshu, Lanjouw P., Murgai, R., & Stern, N. (2013). 'Nonfarm diversification, poverty, economic mobility, and income inequality: A case study in village India. *Agricultural Economics, 44*, 461–473.

Ivanic, M., & Martin, W. (2014). *Short- and long-run impacts of food price changes on poverty, Policy Research Working Paper 7011*. World Bank.

Johnson, H. G. (1967). The possibility of income losses from increased efficiency or factor accumulation in the presence of tariffs. *The Economic Journal*, 151–154.

Kehoe, T. (2005). An Evaluation of the Performance of Applied General Equilibrium Models of the Impact of NAFTA. In T. Kehoe, T. N. Srinivasan, & J. Whalley (Eds.). *Frontiers in Applied General Equilibrium Modeling. In Honor of Herbert Scarf* (pp. 341–377). Cambridge University Press.

Loayza, N., & Raddatz, C. (2010). The composition of growth matters for poverty alleviation. *Journal of Development Economics, 93*, 137–151.

Martin, W., & Alston, J. M. (1997). Producer surplus without apology? *Evaluating Investments in R&D'. Economic Record, 73*(221), 146–158.

Obstfeld, M., & Rogoff, K. (1996). *Foundations of International Macroeconomics*. Cambridge MA: MIT Press.

Ravallion, M., & Datt, G. (1996). How important to the poor is the sectoral composition of economic growth? *World Bank Economic Review, 10*(1), 1–21.

Reimer, J., & Hertel, T. (2004). Estimation of international demand behaviour for use with input–output based data. *Economic Systems Research, 16*(4), 347–366.